

Estimation Error in Latent High-Dimensional Factor Models

Alec Kercheval, Nicholas Gunther, Alex Bernstein, Darwin Yao
Tian Lan, Ethan Lin and Lisa R. Goldberg*

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Abstract

We quantify the error in exposures of a high-dimensional, latent factor model estimated from relatively scarce data. When the number of observations is small, the error may be substantial. We show the error is a sum of two interpretable terms with different characteristics. The out-of-subspace error measures the component of an estimated factor exposure orthogonal to the space spanned by the true exposures, and it can be expressed in terms of observable factor model parameters. The in-subspace error measures the effect of noise in realized factor returns, and it is latent. We conclude our analysis with a three-factor simulation calibrated to the US public equity market. The results measure the dependence of the error and its components on model dimension and sample size. In that simulation, the out-of-subspace component of error is dominant.

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1 Introduction

Factor models are ubiquitous in modern quantitative finance and empirical economics, providing a parsimonious framework to explain the cross-section of returns, manage portfolio risk, and forecast economic outcomes. In the classical setup, a p -vector of observable variables y follows a linear structure

$$y = Bf + z.$$

The product of the factor exposures B and the signal vector f is the systematic component of y , while the idiosyncratic vector z is the component of y not explained by factors. We assume a latent model, in which neither B nor f can be observed. The central object of interest is often the space spanned by the columns of B or, more specifically, the *principal directions*: canonical linear combinations of the exposures B that are identifiable and can be estimated from data. They serve as fundamental building blocks for constructing factor-mimicking portfolios, estimating risk premia, and performing asset pricing tests.

In the language of finance, the variables y often represent security returns, the components of f the factor returns, the columns of B the exposures or sensitivities of the securities to each factor, and the components of z the specific or idiosyncratic returns. In contrast, in engineering or science applications, Bf is often called the signal, with B the factor loading matrix, the components of f the common factors, and z the noise. It's helpful to have both lexicons in mind.

In this article, we quantify and attribute the error in estimated principal directions in the high-dimension, low-sample-size (HL) setting, where parameters are numerous and data are scarce. This setting is commonplace in finance, since market dynamics and non-stationarity of a vast number of securities leave only a limited pool of relevant data available for estimation. Our main result, Theorem 1, is that, with a fixed number of observations, the error has an almost sure limit as $p \rightarrow \infty$ in terms of the model parameters, conditional on the unobserved realizations of f , and it splits into a sum of two interpretable components. The first is the out-of-subspace component of the error, orthogonal to $\text{col}(B)$, which can be estimated from data and serves as a data-driven lower bound for the total error. The second component is the in-subspace error arising from sampling error in the (unobserved) signal vector. This decomposition illuminates how factor estimates deteriorate in high-dimensional, small-sample environments.

Our results have immediate utility for a practitioner fitting a latent factor model to data. For example, we can answer these questions:

- How many observations are required to provide reasonable confidence that estimated factor exposures are accurate enough for the application I have in mind?
- Conversely, for a given number of observations, how large are the errors in estimated factor exposures?
- How accurately have we separated the signal variables from the specific variables?
- How many observations are required to ensure that the latent component of the error in an estimated factor exposure is small enough to neglect, with, say, 95% confidence.

Our results can inform the choice between a simpler model that assumes iid signals f and noise z , versus a more complex model that allows us to access a longer history.

The remainder of the paper is organized as follows. Section 2 provides context for our work with an emphasis on the contributions that we built upon. Section 3 presents the factor model and defines the estimation targets and principal-direction coordinates, which serve as the foundation of our analysis. Section 4 describes the data, scaling, and assumptions. The main theorem is set up in Section 5, stated in Section 6 and proved in Section 7. Section 8 discusses data-driven estimators provided by our analysis. For comparison, the noiseless case is described in section 9. Section 10 presents simulation

results illustrating the theorem. Our work makes use of important results from the mathematical literature, which are tailored to our framework and notation in Appendix A. Tables summarizing the symbols we use are in Appendix B.

2 Context and related literature

Factor models and their close cousins, spiked models (or spiked covariance models) make high dimensional data tractable and separate signal from noise. There is an immense literature on their estimation, although the use of techniques from the high dimension low sample size (HL) regime (Hall et al. 2005, Ahn et al. 2007, Jung et al. 2012, Shen, Shen & Marron 2016, Shen, Shen, Zhu & Marron 2016, Jung et al. 2018, Aoshima et al. 2018) is relatively recent.

A series of articles on James-Stein shrinkage correction of sample eigenvectors serve as precursors to the present article. Goldberg et al. (2022), Shkolnik (2022), Gurdogan & Kercheval (2022), Goldberg & Kercheval (2023), Goldberg et al. (2025), Yoon & Jung (2025) include analysis of the leading sample eigenvector for the simplified case of a one-factor model, with various stationarity and independence assumptions. These articles show that while the leading sample eigenvector is not a consistent estimator of the population counterpart, there is a consistent estimator of the eigenvector *error*. It is this estimator, which emerges from concentration of measure (Talagrand 1995), that allows for James-Stein type corrections to estimated eigenvectors in the HL regime.

The present article may be viewed as an extension of the analysis of estimation error in the leading eigenvector to multiple factors. With more than one factor, there is no longer a consistent estimator of the directional eigenvector error (as shown in Gurdogan & Shkolnik (2026)). Our main theorem breaks down the asymptotic error into a term that can be estimated from data, and a term with an explicit dependence on the distribution of the factor variables. This result may add perspective to the multi-factor James-Stein shrinkage examples in Goldberg et al. (2020), Gurdogan & Shkolnik (2024), Shkolnik et al. (2025).

The closest antecedent to the present analysis is Jung et al. (2012), who study spiked HL models for the case in which the spikes grow linearly in the dimension, which is the relevant case for factor models. They provide a joint unconditional limiting distribution of the sample-population eigenvector angles in terms of lower dimensional Gram matrices similar in spirit to ours. Our factor model context allows us to split the factor and specific parts of the covariance structure, allowing us to decompose the errors into factor-relevant pieces, including a data-driven asymptotic lower bound. We obtain almost sure convergence, rather than convergence in distribution, allowing us a path-by-path analysis. Jung et al. (2012) require the n observations to be independent and identically distributed, but allow cross-sectional weak dependence (ρ -mixing). Our Assumption 1 requires cross-sectional independence of the noise Z conditional on the factor path F , but leaves the joint law unrestricted, allowing factors to be serially dependent and non-stationary, as appropriate to financial data. Neither paper requires the model variables to be Gaussian, sub-Gaussian, sub-exponential, or to belong to any restricted class other than having bounded fourth moments.

3 The factor model and the estimation target

A random p -vector y follows a k -factor linear model

$$y = Bf + z, \tag{1}$$

with $p \gg k$, deterministic loadings $B = B^{(p)} \in \mathbb{R}^{p \times k}$ of rank k , a random signal k -vector f , and a random specific (or idiosyncratic) p -vector $z = z^{(p)}$. Only y is observed. We take f and z to have mean zero and finite second moments, with z uncorrelated with f . Write $\Sigma_f = \mathbb{E}[ff^\top] \in \mathbb{R}^{k \times k}$ (positive

definite) and $\Delta_z = \mathbb{E}[zz^\top] \in \mathbb{R}^{p \times p}$. Then

$$\Sigma_y = B\Sigma_f B^\top + \Delta_z = \Sigma_0 + \Delta_z, \quad \text{where } \Sigma_0 := B\Sigma_f B^\top \text{ (rank } k). \quad (2)$$

Because B and f enter (1) only through their product Bf , and because that product does not have a unique factorization, the components B and f are not separately identifiable. What *is* identifiable is the eigenstructure of the *signal covariance matrix* Σ_0 . Assuming its k nonzero eigenvalues are distinct, the spectral decomposition theorem gives

$$\Sigma_0 = b \Delta_0 b^\top, \quad (3)$$

where $\Delta_0 \in \mathbb{R}^{k \times k}$ is the diagonal matrix of the k leading eigenvalues of Σ_0 in decreasing order and $b = [b_1 \cdots b_k] \in \mathbb{R}^{p \times k}$ has the corresponding orthonormal unit eigenvectors as columns. The b_j s are the *principal directions* of the signal covariance matrix and they are the targets of estimation, with

$$b^\top b = I_k, \quad \text{col}(b) = \text{col}(B) =: \mathcal{B}. \quad (4)$$

Since the columns of b and B form bases for the same k -dimensional subspace $\mathcal{B}$, there is a unique invertible $k \times k$ matrix C for which $b = BC$. Setting

$$\phi = C^{-1}f \in \mathbb{R}^k,$$

we have $Bf = b\phi$ and we can rewrite (1) using principal-direction coordinates:

$$y = b\phi + z. \quad (5)$$

By Lemma 2 of the Appendix, C is computable from the eigenbasis of the $k \times k$ signal Gram $K^{(p)} = \Sigma_f^{1/2} G_B^{(p)} \Sigma_f^{1/2}$, and the covariance matrix $E[\phi\phi^\top]$ of ϕ is equal to Δ_0 , the diagonal matrix of eigenvalues of Σ_0 . Thus the factor contribution $b\phi = \sum_j \phi_j b_j$ decomposes along the orthonormal principal axes b_j with uncorrelated coordinates ϕ_j of variance $(\Delta_0)_{jj}$. We note that the transformation C , and hence the transformed signal k -vector ϕ , depends on p , while the original signal k -vector f does not.

4 Data, scaling, and assumptions

We obtain n observations $y^{(l)}$ determined by n samples $(f^{(l)}, z^{(l)}) \in \mathbb{R}^k \times \mathbb{R}^p$, $l = 1, \dots, n$, of the model (1): each pair satisfies the structure of Section 3 — mean zero, $z^{(l)}$ uncorrelated with $f^{(l)}$, and the *same* factor covariance $\mathbb{E}[f^{(l)} f^{(l)\top}] = \Sigma_f$ — but the samples need not be independent, nor identically distributed. In particular the joint law of the factor path $F = (f^{(1)}, \dots, f^{(n)})$ is unrestricted: the factors may be serially dependent, with Σ_f their common unconditional covariance. The common Σ_f and loadings B are what make the targets b_j of (3) well defined; the structure the proofs require of the specific noise, conditional on the factor path, is collected in Assumptions 1–2 below. The cross-sections are *nested*: for each observation there is a single infinite sequence of specific returns $z_1, z_2, \dots$, and $z^{(p)}$ consists of its first p coordinates. Equivalently, the noise realizations form one infinite array $\{Z_{il} : i \geq 1, 1 \leq l \leq n\}$ from which the $p \times n$ noise matrix at cross-section size p is read off; growing p adds coordinates without changing the law of those already present. (This is the structure under which the strong laws of Proposition 1 apply; see the proof of part (a).)

The observations $y^{(1)}, \dots, y^{(n)}$ are collected as the columns of a $p \times n$ data matrix,

$$Y = Y^{(p,n)} = b\Phi + Z, \quad (6)$$

where $\Phi = \Phi^{(p)} \in \mathbb{R}^{k \times n}$ holds the n realizations of ϕ (equivalently $\Phi = (C^{(p)})^{-1}F$ for the realized factors $F = [f^{(1)}, \dots, f^{(n)}] \in \mathbb{R}^{k \times n}$) and $Z = Z^{(p)} = [z^{(1)}, \dots, z^{(n)}] \in \mathbb{R}^{p \times n}$ the realizations of z . The

asymptotics are taken in p with $n > k$ fixed, conditional on a realization F that does not change with p .

The estimator is the leading sample eigenstructure of the *scaled sample covariance*

$$S^{(p,n)} := \frac{YY^\top}{np} \in \mathbb{R}^{p \times p}, \quad (7)$$

whose j th unit eigenvector we denote $h_j = h_j^{(p,n)}$ (eigenvalue $\theta_j^{(p,n)}$), for $j = 1, \dots, k$, in decreasing order. The $1/p$ scaling keeps the leading eigenvalues $O(1)$: indeed the scaled signal covariance Σ_0/p has leading eigenvalues $\lambda_j^{(p)} := (\Delta_0)_{jj}/p$ and eigenvectors b_j . We compare h_j with its target b_j through $\sin^2 \angle(h_j, b_j)$.

To avoid any angle ambiguity due to sign flips, we adopt the definition

$$\angle(u, v) := \arccos \frac{|\langle u, v \rangle|}{\|u\| \|v\|} \in [0, \frac{\pi}{2}] \quad (8)$$

for any nonzero $u, v \in \mathbb{R}^m$, denoting the acute angle between the lines spanned by u and v . This is the intrinsic notion for eigenvectors, which are determined only up to sign.

Assumption 1 (Specific noise). *Conditional on the factor path $F = (f^{(1)}, \dots, f^{(n)})$, the entries of the noise array $\{Z_{il} : i \geq 1, 1 \leq l \leq n\}$ are mutually independent and mean zero, with conditional variances $\text{Var}(Z_{il} | F) = \delta_{i,l}^2 > 0$ not depending on F , and uniformly bounded conditional fourth moments $\sup_{i,l} \mathbb{E}[Z_{il}^4 | F] \leq \kappa_4 < \infty$ a.s.*

By Jensen's inequality $\delta_{i,l}^2 \leq \sqrt{\kappa_4}$, so $\bar{\delta}^2 := \sup_{i,l} \delta_{i,l}^2 < \infty$.

Remark (A sufficient condition). Assumption 1 holds, in particular, whenever the samples $(f^{(l)}, z^{(l)})$ are independent across l and each satisfies the own-factor condition: conditional on $f^{(l)}$, the entries of $z^{(l)}$ are independent and mean zero with $f^{(l)}$ -free variances $\delta_{i,l}^2 > 0$ and conditional fourth moments bounded by κ_4 .

Assumption 2 (Average specific variance). *For every $l = 1, \dots, n$, $\frac{1}{p} \sum_{i=1}^p \delta_{i,l}^2 \rightarrow \delta^2 > 0$ as $p \rightarrow \infty$, with the same limit δ^2 for each l .*

Assumption 3 (Loading Gram convergence). $\frac{B^\top B}{p} \rightarrow G_B$ for some positive definite $G_B \in \mathbb{R}^{k \times k}$.

Assumption 4 (Population separation). *The $k \times k$ symmetric matrix $\Sigma_f^{1/2} G_B \Sigma_f^{1/2}$ has distinct positive eigenvalues $\lambda_1 > \dots > \lambda_k > 0$. (These are the limiting leading eigenvalues of Σ_0/p , i.e. $\lambda_j^{(p)} \rightarrow \lambda_j$; see Lemma 5.)*

Assumption 5 (Regular event). *The $n \times n$ symmetric matrix $F^\top G_B F/n$ has k distinct positive eigenvalues $\lambda_1^{(n)} > \dots > \lambda_k^{(n)} > 0$.*

Assumptions 1–2 control the noise; 3–4 are population conditions on the signal; 5 a genericity condition on the realized factors.

Assumption 1 constrains the specific noise, not the factors: the joint law of the factor path is unrestricted, so the samples $y^{(l)}$ may be serially dependent through the factors, and need not be identically distributed. Beyond its first two conditional moments, the conditional distribution of Z_{il} may vary with the coordinate i , the observation l , and the whole factor path F , subject to the stated constraints; by the nesting convention preceding (6) it does not vary with p . In particular a volatility regime that differs across the n dates is admissible, provided each cross-sectional average settles at the same level $\delta^2 > 0$, which is the isotropy required by Assumption 2. What Assumption 1 does exclude is feedback

from the specific noise into future factors, Our results can be shown to persist if the cross-sectional independence of Assumption 1 is relaxed to a weak mixing condition; to avoid further complexity we omit the details of this generalization.

Assumption 3 is a prevalence condition implying that for each factor j , a positive limiting fraction of assets have non-negligible exposure to that factor. Assumptions 4 and 5 are generic conditions allowing us to line up the corresponding limiting eigenvalues, as described in the proof. It is straightforward to show that Assumption 5 holds almost surely under any joint law of the path F with a density on $\mathbb{R}^{k \times n}$.

Throughout, statements of the form “conditional on F and almost surely as $p \rightarrow \infty$ ” are to be read as follows: for almost every realization F satisfying Assumption 5, the stated convergence holds with probability one under the conditional law of the noise array $\{Z_{il}\}$ given F — the law directly regulated by Assumption 1. That Assumption fixes the conditional mean and variance of each Z_{il} but permits the rest of its conditional distribution to depend on the path F (and on i and l), so the noise law may genuinely vary with the realization being conditioned on.

5 Two small Gram matrices

The growing ambient dimension is handled by passing to fixed-size Gram matrices.

The $n \times n$ dual. The matrices $YY^\top/(np) \in \mathbb{R}^{p \times p}$ and

$$W^{(p,n)} := \frac{Y^\top Y}{np} \in \mathbb{R}^{n \times n} \quad (9)$$

share their nonzero eigenvalues (Lemma 1); $W^{(p,n)}$ has the same fixed size n at every p . Its (l, m) entry is the scaled inner product $\langle Y_{\cdot l}, Y_{\cdot m} \rangle / (np)$ of two observations. Substituting $Y = b\Phi + Z$ and using $b^\top b = I$, this converges (Proposition 1) to

$$W^{(n)} := \underbrace{\frac{F^\top G_B F}{n}}_{=: W^{(n),0}} + \frac{\delta^2}{n} I_n, \quad (10)$$

a signal Gram $W^{(n),0}$ plus an *isotropic* noise term: independent noise across the p coordinates averages to the same variance δ^2/n at each observation. The noiseless part $W^{(n),0}$ has rank k ; write its j th eigenpair as $(\lambda_j^{(n)}, w_j^{(n)})$, with the $\lambda_j^{(n)}$ of Assumption 5. Adding $(\delta^2/n)I_n$ shifts every eigenvalue by δ^2/n and fixes the eigenvectors, so the top k eigenpairs of $W^{(n)}$ are $(\lambda_j^{(n)} + \delta^2/n, w_j^{(n)})$, separated from the remaining eigenvalue δ^2/n (multiplicity $n - k$) by the gap $\lambda_k^{(n)} > 0$.

The $k \times k$ realized signal Gram in principal coordinates. With the noiseless data $b\Phi$, the scaled sample covariance is $b\Phi\Phi^\top b^\top/(np) = bN^{(p,n)}b^\top$, where

$$N^{(p,n)} := \frac{\Phi\Phi^\top}{np} \in \mathbb{R}^{k \times k} \quad (11)$$

is the realized signal covariance read in principal coordinates. Defining $W^{(p,n),0} := F^\top (B^\top B/p)F/n$, the dual of $N^{(p,n)}$ is

$$\Phi^\top \Phi / (np) = W^{(p,n),0} \rightarrow W^{(n),0} \in \mathbb{R}^{n \times n} \quad (12)$$

by Lemma 5, and $N^{(p,n)}$ and $W^{(p,n),0}$ share the nonzero spectrum $\{\lambda_j^{(p,n)}\}$ by Lemma 1. Lemma 5 establishes the limit

$$N^{(p,n)} \rightarrow N^{(n)}, \quad N^{(n)} \text{ has } j\text{th unit eigenpair } (\lambda_j^{(n)}, \nu_j^{(n)}), \quad (13)$$

and the duality link $\sqrt{n\lambda_j^{(n)}} \nu_j^{(n)} = \bar{\Phi}^\infty w_j^{(n)}$ connecting eigenvectors of $W^{(n)}$ and $N^{(n)}$, where $\bar{\Phi}^\infty = \lim_{p \rightarrow \infty} \Phi/\sqrt{p}$. Lemma 5 also provides explicit expressions for $\bar{\Phi}^\infty$ and $N^{(n)}$ in terms of the model parameters.

The direction $\nu_j^{(n)} \in \mathbb{R}^k$ is the limiting realized j th principal signal direction *in principal coordinates*. Were the factor covariance known exactly ($n = \infty$, $FF^\top/n = \Sigma_f$), one would have $N^{(n)} = \text{diag}(\lambda_1, \dots, \lambda_k)$ (Lemma 5), whose eigenvectors are the axes e_j . Finite n sample error rotates $\nu_j^{(n)}$ off the axis e_j , and that rotation is the in-subspace error below.

6 Main theorem

Theorem 1. *Under Assumptions 1–5, conditional on F and almost surely as $p \rightarrow \infty$, for each $j \in \{1, \dots, k\}$,*

$$\sin^2 \angle(h_j, b_j) \longrightarrow \underbrace{\frac{\delta^2}{n\lambda_j^{(n)} + \delta^2}}_{\text{out-of-subspace floor}} + \underbrace{\frac{n\lambda_j^{(n)}}{n\lambda_j^{(n)} + \delta^2} \sin^2 \angle(\nu_j^{(n)}, e_j)}_{\text{in-subspace rotation}}, \quad (14)$$

where $(\lambda_j^{(n)}, \nu_j^{(n)})$ is the j th eigenpair of $N^{(n)}$, defined in the previous section. The target $b_j = b_j^{(p)}$ and the estimate $h_j = h_j^{(p,n)}$ are both evaluated at cross-section size p ; the angle is formed at each p .

The floor is the noise's share of the direction- j variance and persists as $p \rightarrow \infty$; the rotation measures how far the realized principal direction $\nu_j^{(n)}$ has turned from its axis e_j . Both terms vanish as $n \rightarrow \infty$ ($\lambda_j^{(n)} \rightarrow \lambda_j$ and $\nu_j^{(n)} \rightarrow e_j$; Lemma 5(iv)).

As the proof shows, the out-of-subspace floor is the limit of $\sin^2 \angle(h_j, \mathcal{B})$, the squared sine of the angle between h_j and $\mathcal{B} = \text{col}(B)$; the in-subspace rotation is the limit of the remaining angular error between the projection of h_j onto $\mathcal{B}$ and the target direction b_j .

Remark (Theorem 1 in terms of signal-to-noise ratios). Using the realized signal-to-noise ratio, $\text{SNR}_j^{(n)} = n\lambda_j^{(n)}/\delta^2$, Theorem 1 can be rewritten as

$$\sin^2 \angle(h_j, b_j) \longrightarrow \frac{1}{1 + \text{SNR}_j^{(n)}} + \frac{\text{SNR}_j^{(n)}}{1 + \text{SNR}_j^{(n)}} \sin^2 \angle(\nu_j^{(n)}, e_j). \quad (15)$$

This formulation makes the role of the finite-sample signal-to-noise ratio explicit: the out-of-subspace floor is the fraction of variance attributable to noise, while the in-subspace rotation is weighted by the fraction attributable to signal. The realized signal-to-noise ratio $\text{SNR}_j^{(n)}$ itself converges to the population signal-to-noise ratio $\text{SNR}_j = n\lambda_j/\delta^2$ as $n \rightarrow \infty$, and the quality of this approximation as a function of n governs the accuracy of the limiting error.

7 Proof of Theorem 1

All limits are almost sure as $p \rightarrow \infty$ with n, k fixed; eigenvectors are taken up to sign, and every quantity in (14) is sign-free.

Our proof relies on principal direction coordinates for the factor contribution, which gives us the model representation

$$y = b\phi + z. \quad (16)$$

This provides us with a coordinate mapping for the factor loading subspace $\mathcal{B} = \text{col}(b)$, given by

$$\Gamma : \mathbb{R}^k \rightarrow \mathcal{B}, \quad \Gamma(\xi) = b\xi \in \mathcal{B}, \quad (17)$$

and so the principal-direction coordinate of any $u \in \mathcal{B}$ is $\Gamma^{-1}(u) = b^\top u \in \mathbb{R}^k$. Since b has orthonormal columns, Γ is an isometry from $\mathbb{R}^k$ to $\mathcal{B}$ with its inherited inner product as a subspace of $\mathbb{R}^p$. The factor term $b\phi = \sum_j \phi_j b_j$ is a decomposition of $\Gamma(\phi) \in \mathcal{B}$ along the orthonormal principal-coordinate directions b_j with uncorrelated coordinates ϕ_j of variance $(\Delta_0)_{jj}$ (Lemma 2).

In these principal-direction coordinates on $\mathbb{R}^k = \Gamma^{-1}(\mathcal{B})$, this means that the estimation targets b_j are simply the coordinate axes $e_j = \Gamma^{-1}(b_j)$ of $\mathbb{R}^k$. This explains the appearance of e_j in the statement of the theorem.

We now proceed to the steps of the proof. Fix j . Let $\Pi := bb^\top$ be the orthogonal projector onto $\mathcal{B}$ and $\Pi^\perp := I - \Pi$, and write $\|\cdot\|_F$ for the Frobenius norm.

Step 0: an exact angular split. Decompose $h_j = \Pi^\perp h_j + \Pi h_j$ into its components orthogonal to and inside $\mathcal{B}$. Since $\langle \Pi^\perp h_j, b_j \rangle = 0$ (as $b_j \in \mathcal{B}$),

$$\cos^2 \angle(h_j, b_j) = \langle h_j, b_j \rangle^2 = \langle \Pi h_j, b_j \rangle^2 = \|\Pi h_j\|^2 \cos^2 \angle\left(\frac{\Pi h_j}{\|\Pi h_j\|}, b_j\right), \quad (18)$$

and using $\|\Pi^\perp h_j\|^2 = 1 - \|\Pi h_j\|^2$ and $\sin^2 = 1 - \cos^2$,

$$\sin^2 \angle(h_j, b_j) = \|\Pi^\perp h_j\|^2 + \|\Pi h_j\|^2 \sin^2 \angle\left(\frac{\Pi h_j}{\|\Pi h_j\|}, b_j\right), \quad (19)$$

an identity valid at every p (the second summand is set to 0 if $\Pi h_j = 0$). The proof evaluates the three limits $\|\Pi^\perp h_j\|^2$, $\|\Pi h_j\|^2$, and the inner angle.

Step 1: reduce to the fixed dual. For convenience, let $A := Y/\sqrt{np}$, so $S^{(p,n)} = AA^\top$ and $W^{(p,n)} = A^\top A$ share the eigenvalue $\theta_j^{(p,n)}$. We first note that the finite- p objects below are eventually well defined: by Proposition 1(c) and Weyl's inequality, every eigenvalue of $W^{(p,n)}$ converges to the corresponding eigenvalue of $W^{(n)}$, and the $k+1$ limits $\lambda_1^{(n)} + \delta^2/n > \dots > \lambda_k^{(n)} + \delta^2/n > \delta^2/n$ are distinct and positive (Assumption 5); hence, on the a.s. event of the proposition and for all sufficiently large p , the top k eigenvalues $\theta_1^{(p,n)} > \dots > \theta_k^{(p,n)} > 0$ of $W^{(p,n)}$ are simple and strictly positive. All finite- p statements below are understood for such p ; in particular Lemma 1 applies to A and the reconstruction (20) is well defined. Recall $w_j^{(p,n)} \in \mathbb{R}^n$ is the j th unit eigenvector of $W^{(p,n)}$. Gram duality (Lemma 1) reconstructs the ambient eigenvector from this fixed-size one:

$$h_j = \frac{A w_j^{(p,n)}}{\sqrt{\theta_j^{(p,n)}}} = \frac{Y w_j^{(p,n)}}{\sqrt{np} \sqrt{\theta_j^{(p,n)}}} = \frac{b \Phi w_j^{(p,n)}}{\sqrt{p} \sqrt{n \theta_j^{(p,n)}}} + \frac{Z w_j^{(p,n)}}{\sqrt{p} \sqrt{n \theta_j^{(p,n)}}}. \quad (20)$$

Projecting with $\Pi^\perp$ and Π : $\Pi^\perp$ annihilates $b \Phi w_j^{(p,n)} \in \mathcal{B}$, while $\Pi b = b$, so the signal part belongs to $\mathcal{B}$ and its principal coordinate $\xi_j^{(p,n)} \in \mathbb{R}^k$ is obtained by multiplying on the left by $b^\top$. This gives the exact finite- p identities

$$\Pi^\perp h_j = \frac{\Pi^\perp Z w_j^{(p,n)}}{\sqrt{p} \sqrt{n \theta_j^{(p,n)}}}, \quad \Pi h_j = b \xi_j^{(p,n)} + \frac{\Pi Z w_j^{(p,n)}}{\sqrt{p} \sqrt{n \theta_j^{(p,n)}}}, \quad \text{where } \xi_j^{(p,n)} := \frac{\bar{\Phi} w_j^{(p,n)}}{\sqrt{n} \sqrt{\theta_j^{(p,n)}}} \quad (21)$$

and $\bar{\Phi} := \Phi/\sqrt{p} \in \mathbb{R}^{k \times n}$. All p -dependence is now carried by the noise Z and the fixed-size quantities $\theta_j^{(p,n)}$, $w_j^{(p,n)}$, $\bar{\Phi}$, governed by the limits (Proposition 1 and Lemma 5)

$$\frac{Z^\top Z}{p} \rightarrow \delta^2 I_n, \quad \|\Pi Z\|_F = o(\sqrt{p}), \quad \theta_j^{(p,n)} \rightarrow \lambda_j^{(n)} + \frac{\delta^2}{n}, \quad w_j^{(p,n)} \rightarrow w_j^{(n)}, \quad \bar{\Phi} \rightarrow \bar{\Phi}^\infty. \quad (22)$$

Step 2: out-of-subspace floor, $\|\Pi^\perp h_j\|^2 \rightarrow \delta^2/(n\lambda_j^{(n)} + \delta^2)$. By (21) and $\|\Pi^\perp v\|^2 = \|v\|^2 - \|\Pi v\|^2$ with $v = Zw_j^{(p,n)}$,

$$\|\Pi^\perp h_j\|^2 = \frac{\|\Pi^\perp Zw_j^{(p,n)}\|^2}{pn\theta_j^{(p,n)}} = \frac{1}{n\theta_j^{(p,n)}} \left((w_j^{(p,n)})^\top \frac{Z^\top Z}{p} w_j^{(p,n)} - \frac{\|\Pi Zw_j^{(p,n)}\|^2}{p} \right). \quad (23)$$

Since $\|w_j^{(p,n)}\| = 1$ and $Z^\top Z/p \rightarrow \delta^2 I_n$, the first bracketed term $\rightarrow \delta^2$ (no limit of $w_j^{(p,n)}$ is needed); the second is $\leq \|\Pi Z\|_F^2/p = o(1)$ by (22). With $n\theta_j^{(p,n)} \rightarrow n\lambda_j^{(n)} + \delta^2$,

$$\|\Pi^\perp h_j\|^2 \rightarrow \frac{\delta^2}{n\lambda_j^{(n)} + \delta^2}. \quad (24)$$

This is a variance partition: of the total direction- j variance $\lambda_j^{(n)} + \delta^2/n$ (the dual eigenvalue), the share δ^2/n is noise.

Step 3: in-subspace mass, $\|\Pi h_j\|^2 \rightarrow (\kappa_j^{(n)})^2$. The noise term of Πh_j in (21) has norm at most $\|\Pi Z\|_F/(\sqrt{np}\sqrt{\theta_j^{(p,n)}})$, which is $o(\sqrt{p})/\Theta(\sqrt{np}) = o(1)$; so

$$\Pi h_j = b\xi_j^{(p,n)} + o(1). \quad (25)$$

Using the limits $\bar{\Phi} \rightarrow \bar{\Phi}^\infty$, $w_j^{(p,n)} \rightarrow w_j^{(n)}$, and $\theta_j^{(p,n)} \rightarrow \lambda_j^{(n)} + \delta^2/n$, together with the duality link $\bar{\Phi}^\infty w_j^{(n)} = \sqrt{n\lambda_j^{(n)}} \nu_j^{(n)}$ of Lemma 5,

$$\xi_j^{(p,n)} = \frac{\bar{\Phi} w_j^{(p,n)}}{\sqrt{n} \sqrt{\theta_j^{(p,n)}}} \rightarrow \frac{\bar{\Phi}^\infty w_j^{(n)}}{\sqrt{n\lambda_j^{(n)} + \delta^2}} = \frac{\sqrt{n\lambda_j^{(n)}}}{\sqrt{n\lambda_j^{(n)} + \delta^2}} \nu_j^{(n)} =: \kappa_j^{(n)} \nu_j^{(n)}, \quad (26)$$

with $\kappa_j^{(n)} := \sqrt{n\lambda_j^{(n)}/(n\lambda_j^{(n)} + \delta^2)} \in (0, 1)$. (Note here is where the limit of $w_j^{(p,n)}$ is needed.) Because b is an isometry and $\|\nu_j^{(n)}\| = 1$,

$$\|\Pi h_j\|^2 = \|\xi_j^{(p,n)}\|^2 + o(1) \rightarrow (\kappa_j^{(n)})^2 = \frac{n\lambda_j^{(n)}}{n\lambda_j^{(n)} + \delta^2}. \quad (27)$$

In particular $\kappa_j^{(n)} > 0$, so the normalized projection in (19) is well defined for large p .

Step 4: in-subspace rotation, $\sin^2 \angle(\Pi h_j / \|\Pi h_j\|, b_j) \rightarrow \sin^2 \angle(\nu_j^{(n)}, e_j)$. The target coordinate is e_j exactly. Using $\Pi h_j = b\xi_j^{(p,n)} + o(1)$, the isometry b , (26), and $\|\Pi h_j\| \rightarrow \kappa_j^{(n)} > 0$,

$$\cos \angle\left(\frac{\Pi h_j}{\|\Pi h_j\|}, b_j\right) = \frac{\langle b\xi_j^{(p,n)}, be_j \rangle + o(1)}{\|\Pi h_j\|} = \frac{\langle \xi_j^{(p,n)}, e_j \rangle + o(1)}{\|\Pi h_j\|} \rightarrow \frac{\kappa_j^{(n)}(\nu_j^{(n)})_j}{\kappa_j^{(n)}} = (\nu_j^{(n)})_j = \nu_j^{(n)\top} e_j, \quad (28)$$

the j th component of $\nu_j^{(n)}$. Hence

$$\sin^2 \angle\left(\frac{\Pi h_j}{\|\Pi h_j\|}, b_j\right) \rightarrow 1 - (\nu_j^{(n)\top} e_j)^2 = \sin^2 \angle(\nu_j^{(n)}, e_j). \quad (29)$$

Conclusion. Substituting (24), (27), (29) into the exact split (19) yields (14). $\square$

8 Data-driven estimators

The decomposition of Theorem 1 splits the limiting error into estimable and un-estimable components. The out-of-subspace floor is written through the population quantities δ^2 and $\lambda_j^{(n)}$, neither of which is observed, but are nonetheless estimable: as we describe below, the floor equals an asymptotic ratio of sample eigenvalues of $S^{(p,n)}$. This means an approximate lower bound on the limiting total alignment error can be read off from the data alone.

The second term in the error, the limiting in-subspace rotation, depends on the unobserved F and is not consistently estimable from observations of y alone. This conclusion is supported by the impossibility theorem of Gurdogan & Shkolnik (2024) and Gurdogan & Shkolnik (2026), which states that the pairwise angles in $H^\top B$ cannot be consistently estimated from data.

Recall that $\theta_1^{(p,n)} \geq \dots \geq \theta_n^{(p,n)}$ denote the eigenvalues of the dual $W^{(p,n)} = Y^\top Y/(np)$; by Gram duality (Lemma 1) these are exactly the n largest eigenvalues of the scaled sample covariance $S^{(p,n)} = YY^\top/(np)$, the remaining $p-n$ being zero. Since $n > k$, we may form the average of the $n-k$ trailing (“bulk”) sample eigenvalues,

$$\ell^{(p,n)} := \frac{1}{n-k} \sum_{i=k+1}^n \theta_i^{(p,n)} = \frac{1}{n-k} \left(\text{tr } W^{(p,n)} - \sum_{i=1}^k \theta_i^{(p,n)} \right), \quad (30)$$

the second form using $\text{tr } W^{(p,n)} = \|Y\|_F^2/(np)$ and requiring only the top k eigenvalues explicitly. Both $\ell^{(p,n)}$ and $\theta_j^{(p,n)}$ are computable from the data (given the factor count k).

Corollary 1 (Observable floor). *Under Assumptions 1–5, conditional on F and almost surely as $p \rightarrow \infty$,*

$$\ell^{(p,n)} \rightarrow \frac{\delta^2}{n}, \quad \theta_j^{(p,n)} \rightarrow \lambda_j^{(n)} + \frac{\delta^2}{n} > 0 \quad (j = 1, \dots, k), \quad (31)$$

and hence, for each $j \in \{1, \dots, k\}$,

$$\frac{\ell^{(p,n)}}{\theta_j^{(p,n)}} \rightarrow \frac{\delta^2}{n\lambda_j^{(n)} + \delta^2} = \lim_{p \rightarrow \infty} \|\Pi^\perp h_j\|^2, \quad (32)$$

the out-of-subspace floor of Theorem 1. Therefore

$$\lim_{p \rightarrow \infty} \frac{\ell^{(p,n)}}{\theta_j^{(p,n)}} = \frac{\delta^2}{n\lambda_j^{(n)} + \delta^2} \leq \lim_{p \rightarrow \infty} \sin^2 \angle(h_j, b_j). \quad (33)$$

This means $\ell^{(p,n)}/\theta_j^{(p,n)}$ is a strongly consistent, fully observable estimator of a lower bound on $\sin^2 \angle(h_j, b_j)$ for large p , with equality in (33) precisely when the in-subspace rotation $\sin^2 \angle(\nu_j^{(n)}, e_j)$ vanishes.

Proof. By Proposition 1(c), $W^{(p,n)} \rightarrow W^{(n)}$ in spectral norm, a.s. The limit $W^{(n)} = F^\top G_B F/n + (\delta^2/n)I_n$ of (10) has top k eigenvalues $\lambda_j^{(n)} + \delta^2/n$ and smallest eigenvalue δ^2/n with multiplicity $n-k$: the rank- k signal Gram $F^\top G_B F/n$ contributes $\lambda_j^{(n)}$ on its range and 0 on the orthogonal complement, and the isotropic shift adds δ^2/n throughout (the discussion following (10)). Weyl’s inequality gives, for every i , $|\theta_i^{(p,n)} - \lambda_i(W^{(n)})| \leq \|W^{(p,n)} - W^{(n)}\| \rightarrow 0$; in particular $\theta_i^{(p,n)} \rightarrow \delta^2/n$ for each $i \in \{k+1, \dots, n\}$, and averaging these $n-k$ limits yields $\ell^{(p,n)} \rightarrow \delta^2/n$. The convergence

$\theta_j^{(p,n)} \rightarrow \lambda_j^{(n)} + \delta^2/n > 0$ for $j = 1, \dots, k$ is Proposition 1(d). As the denominator limit is strictly positive, (31) gives

$$\frac{\ell^{(p,n)}}{\theta_j^{(p,n)}} \rightarrow \frac{\delta^2/n}{\lambda_j^{(n)} + \delta^2/n} = \frac{\delta^2}{n\lambda_j^{(n)} + \delta^2},$$

which equals $\lim_{p \rightarrow \infty} \|\Pi^\perp h_j\|^2$ by (24), proving (32). Finally (33) is immediate from (14), whose in-subspace summand is nonnegative and vanishes iff $\sin^2 \angle(\nu_j^{(n)}, e_j) = 0$. $\square$

Remark (Observable signal-to-noise ratio and variance shares). The two limits in (31) also make the realized signal-to-noise ratio of (15) observable:

$$\frac{\theta_j^{(p,n)}}{\ell^{(p,n)}} - 1 \rightarrow \frac{n\lambda_j^{(n)}}{\delta^2} = \text{SNR}_j^{(n)}, \quad (34)$$

so the SNR-form weights are estimable directly from the sample spectrum, without separately identifying δ^2 or $\lambda_j^{(n)}$. Complementarily, $1 - \ell^{(p,n)}/\theta_j^{(p,n)} \rightarrow n\lambda_j^{(n)}/(n\lambda_j^{(n)} + \delta^2) = (\kappa_j^{(n)})^2$, the in-subspace variance share $\lim_{p \rightarrow \infty} \|\Pi h_j\|^2$ of (27). Thus the observable ratio $\ell^{(p,n)}/\theta_j^{(p,n)}$ and its complement recover the noise and signal shares of the direction- j variance partition.

Aggregating the per-direction floors over j yields a subspace-level statement. Let $H := [h_1 \ \dots \ h_k] \in \mathbb{R}^{p \times k}$ collect the top k sample eigenvectors, and let $P_H := HH^\top$ and $P_{\mathcal{B}} := \Pi = bb^\top$ be the orthogonal projectors onto $\text{col}(H)$ and $\mathcal{B}$.

Corollary 2 (Subspace error). *Under Assumptions 1–5, conditional on F and almost surely as $p \rightarrow \infty$,*

$$\frac{1}{2} \|P_H - P_{\mathcal{B}}\|_F^2 = \sum_{j=1}^k \sin^2 \angle(h_j, \mathcal{B}) \rightarrow \sum_{j=1}^k \frac{\delta^2}{n\lambda_j^{(n)} + \delta^2}, \quad (35)$$

the sum of the out-of-subspace floors of Theorem 1. The middle expression also equals the sum of the squared sines of the k principal angles between $\text{col}(H)$ and $\mathcal{B}$. By Corollary 1, the limit is consistently estimated by the observable statistic $\sum_{j=1}^k \ell^{(p,n)}/\theta_j^{(p,n)}$.

Proof. Since $\sin^2 \angle(h_j, \mathcal{B}) = \|\Pi^\perp h_j\|^2$, the limit in (35) is the sum over $j = 1, \dots, k$ of the limits (24) established in Step 2 of the proof of Theorem 1. For the first equality, expand $\|P_H - P_{\mathcal{B}}\|_F^2 = \text{tr } P_H + \text{tr } P_{\mathcal{B}} - 2 \text{tr}(P_H P_{\mathcal{B}}) = 2k - 2 \sum_j h_j^\top \Pi h_j = 2 \sum_j (1 - \|\Pi h_j\|^2) = 2 \sum_j \|\Pi^\perp h_j\|^2$, using that H and b each have k orthonormal columns. For the principal-angle identification: the cosines of the principal angles between the k -dimensional subspaces $\text{col}(H)$ and $\mathcal{B}$ are the singular values $\sigma_1, \dots, \sigma_k$ of $b^\top H$, so $\sum_i \sin^2 \theta_i = k - \sum_i \sigma_i^2 = k - \|b^\top H\|_F^2 = \sum_j (1 - \|\Pi h_j\|^2)$, the same quantity. The final claim combines (35) with (32) summed over j . $\square$

Remark (The rotation is invisible at the subspace level). The in-subspace rotation of Theorem 1 does not appear in (35): rotating the realized directions *within* $\mathcal{B}$ moves the individual eigenvectors but not the subspace they generate, so subspace-level metrics see only the floors. In particular, since each floor is strictly positive, the limit in (35) is strictly positive: the sample subspace $\text{col}(H)$ is not a consistent estimator of $\mathcal{B}$ in Frobenius (hence operator) norm, however large p . Weaker notions of subspace convergence (e.g., convergence of the bilinear forms $u^\top P_H v$ along suitable sequences of test directions) are not ruled out by (35); we do not pursue them here.

9 The noiseless case

Switching the noise off isolates the rotation and makes the error formula exact at finite p .

Corollary 3. *Let $h_j^{(p,n),0}$ be the j th unit eigenvector of the noiseless scaled sample covariance $S^{(p,n),0} = b\Phi\Phi^\top b^\top/(np)$, and let $(\lambda_j^{(p,n)}, \nu_j^{(p,n)})$ be the j th eigenpair of $N^{(p,n)} = \Phi\Phi^\top/(np) \in \mathbb{R}^{k \times k}$. For every p at which these are simple,*

$$\sin^2 \angle(h_j^{(p,n),0}, b_j) = \sin^2 \angle(\nu_j^{(p,n)}, e_j). \quad (36)$$

Proof. Run Step 1 with $Z = 0$. The dual of $S^{(p,n),0}$ is $W^{(p,n),0} = \Phi^\top \Phi/(np)$, with j th eigenpair $(\lambda_j^{(p,n)}, w_j^{(p,n),0})$; Gram duality applied to $\bar{\Phi}/\sqrt{n}$ gives $\bar{\Phi}w_j^{(p,n),0} = \sqrt{n\lambda_j^{(p,n)}} \nu_j^{(p,n)}$. The reconstruction (20) then reads

$$h_j^{(p,n),0} = \frac{b\Phi w_j^{(p,n),0}}{\sqrt{np} \sqrt{\lambda_j^{(p,n)}}} = b \frac{\bar{\Phi}w_j^{(p,n),0}}{\sqrt{n} \sqrt{\lambda_j^{(p,n)}}} = b \nu_j^{(p,n)}, \quad (37)$$

so the principal coordinate of $h_j^{(p,n),0}$ is exactly $\nu_j^{(p,n)}$ and that of b_j is e_j . The principal coordinate isometry gives $\langle h_j^{(p,n),0}, b_j \rangle = (\nu_j^{(p,n)})_j$, and as all four vectors are unit, (36) follows. $\square$

Remark (Comparison: switching on the noise). Corollary 3 and Theorem 1 exhibit the limit transparently. As $p \rightarrow \infty$ the right side of (36) converges to $\sin^2 \angle(\nu_j^{(n)}, e_j)$ (Lemma 5(ii), $N^{(p,n)} \rightarrow N^{(n)}$, with the eigenvector convergence from Lemma 4 and Assumption 5). This limiting value coincides with the formula of Theorem 1 evaluated at the limit $\delta^2 \rightarrow 0$. Switching on the noise produces the out-of-subspace term $\delta^2/(n\lambda_j^{(n)} + \delta^2)$, rescales the surviving in-subspace mass by $(\kappa_j^{(n)})^2 = n\lambda_j^{(n)}/(n\lambda_j^{(n)} + \delta^2)$, and — since the scalar shift $(\delta^2/n)I_n$ in $W^{(n)}$ does not move the eigenvectors $w_j^{(n)}$ — leaves the in-subspace *direction* $\nu_j^{(n)}$ and target e_j untouched.

Remark (Structure of the two terms). We call the out-of-subspace term $\delta^2/(n\lambda_j^{(n)} + \delta^2)$ the *noise floor*: it is determined entirely by $\text{SNR}_j^{(n)}$, is a strictly positive lower bound on $\sin^2 \angle(h_j, b_j)$ that persists in the limit, and cannot be reduced by collecting more variables (increasing p). This irreducibility is the sense in which it is a “floor.” It vanishes only as $n \rightarrow \infty$ or $\delta^2 \rightarrow 0$.

The in-subspace term $\sin^2 \angle(\nu_j^{(n)}, e_j)$ (the *rotation*) measures how far the realized principal direction $\nu_j^{(n)}$ has rotated from its coordinate axis e_j . Since $FF^\top/n \rightarrow \Sigma_f$ as $n \rightarrow \infty$, we have $N^{(n)} \rightarrow \text{diag}(\lambda_1, \dots, \lambda_k)$, so $\nu_j^{(n)} \rightarrow e_j$ and this term vanishes in the large- n limit. In the noiseless limit $\delta^2 \rightarrow 0$, however, the rotation term survives with weight 1.

Equivalently, $1 - (\kappa_j^{(n)})^2 = \delta^2/(n\lambda_j^{(n)} + \delta^2)$ is the squared norm *lost* when projecting h_j onto $\mathcal{B}$: the limit $\kappa_j^{(n)} = \lim_p \|\Pi h_j\| < 1$ measures how far h_j falls short of lying in the signal subspace, and the floor is exactly $1 - (\kappa_j^{(n)})^2$. The in-subspace term $\sin^2 \angle(\nu_j^{(n)}, e_j)$ measures the remaining angular error within $\mathcal{B}$.

Remark (Special cases). When $k = 1$, the rotation term is trivially zero and the formula reduces to $\sin^2 \angle(h_1, b_1) \rightarrow \delta^2/(n\lambda_1^{(n)} + \delta^2)$, recovering the result in Goldberg et al. (2022).

10 Simulation

Guided by the empirical analysis in Bayraktar et al. (2014), we calibrate a three-factor instance of the model specified in (1) to the US public equity market. We assume factor and specific returns are

independently t -distributed over time with mean 0, and 5 and 6 degrees of freedom. Their annualized volatilities are in Table 1.

Table 1: Annualized factor and specific volatilities

	Factor 1	Factor 2	Factor 3	Specific
Volatility	0.16	0.08	0.06	0.40

We set the columns of B , the raw exposures to the factors, using draws from normal distributions $N(1, 0.25)$, $N(0, 1)$ and $N(0, 1)$, and fix them for the entire experiment. The resulting limiting Gram matrix G_B is:

$$G_B = \begin{pmatrix} 1.25 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

In a first experiment, we examine the rate at which finite- p results converge to the asymptotic limit guaranteed by Theorem 1, with the number of variables p increasing from 100 to 50000. We ran 1000 simulation paths based on $n = 63$ observations, three months worth of daily data. This horizon may be short enough to reasonably allow an assumption of stationarity much of the time. A path is characterized by time series of 63 3-vectors of factor returns and 63 p -vectors of specific returns, which are nested as p grows.

The top panel of Figure 1 show finite- p errors averaged over 1000 paths, $\sin^2 \angle(h_j, b_j)$, indicated by black dots. Blue bars are average asymptotic out-of-subspace components of the error, while orange bars are in-subspace components, the summands on the right hand side of (14). Translating from units of sine-squared to angles for $p = 500$ the errors $\angle(h_j, b_j)$ are 17.1° , 39.2° and 49.5° on average for factors 1, 2 and 3. The average out-of-subspace component accounts for 86% of the asymptotic error in the current calibration.

In the bottom panel of Figure 1, we show path-by-path differences between the finite- p error—the angle between the true and estimated factor exposure based on $n = 63$ observations—and the asymptotic limit, the right hand side of (14). In our experiment, estimates for factor 1 are relatively accurate even for $p = 100$. Estimates for factors 2 and 3 require as many as $p = 1000$ variables to be relatively accurate.

In a second experiment, we examine the effect of increasing the number of observations n from 20 to 250, with the number of variables p fixed at 3000. This is roughly the number of securities in the Russell 3000 Index. As in the first experiment, results are based on 1000 paths.

As in the first experiment, results in the top panel of Figure 2 are averages over the 1000 paths, with black dots representing finite- p estimates of the error and blue and orange bars representing out-of-subspace and in-subspace components of the asymptotic error from the right-hand-side of (14). Translating from units of sine-squared to angles, average errors decreased from 30.33° to 8.48° for factor 1, from 56.28° to 19.24° for factor 2, and from 65.75° to 24.45° for factor 3. The percentage of total error accounted for by the average out-of-subspace component of the asymptotic error varies slightly with n for all three factors. The percentages are shown in Table 2.

The ranges of path-by-path differences between the finite- p errors and the asymptotic limits shown in the bottom panel of Figure 2 decrease steadily for all three factors as n increases. There are positive biases in the errors for factors 2 and 3. The vertical scale of the bottom panel in Figure 2 is smaller than in Figure 1. The error remains non-negligible even at $n = 250$; the bias in the errors is positive.

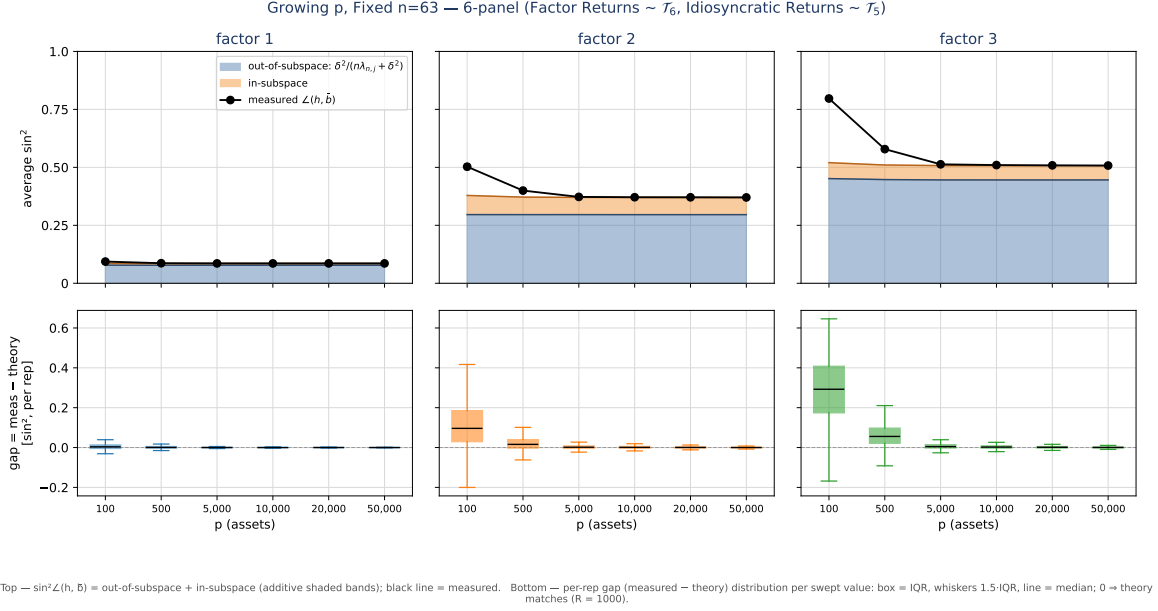


Figure 1: Errors between true and estimated normalized factor exposures based on $n = 63$ observations of p variables. Top panel: Black dots are finite p errors averaged over 1000 paths. Blue and orange bars are average out-of-subspace and in-subspace asymptotic limits. Bottom panel: Path-by-path differences between the finite p angle between the true and estimated normalized factor exposures estimated from $n = 63$ observations and the asymptotic limits.

Table 2: Out-of-subspace share (% of predicted error), fixed $p = 3000$, growing n .

n	20	30	45	60	63	90	120	180	250
factor									
1	88.42	88.69	89.59	89.97	90.04	89.98	91.41	91.66	91.67
2	84.88	82.60	81.87	81.08	80.96	80.14	80.74	81.04	83.19
3	92.68	90.55	89.61	88.99	88.91	88.18	88.37	88.59	90.03

A Supporting Lemmas

Lemma 1 (Gram duality). *Let A be a nonzero matrix such that $AA^\top$ has simple nonzero eigenvalues. If (λ, v) is the j th eigenpair of $AA^\top$, then $(\lambda, A^\top v / \sqrt{\lambda})$ is the j th eigenpair of $A^\top A$, and conversely.*

Proof. From $AA^\top v = \lambda v$ with $\lambda > 0$, set $u := A^\top v / \sqrt{\lambda}$. Then $A^\top A u = A^\top (AA^\top v) / \sqrt{\lambda} = \lambda u$, and $\|u\|^2 = v^\top AA^\top v / \lambda = \|v\|^2 = 1$. The maps $v \mapsto A^\top v / \sqrt{\lambda}$ and $u \mapsto Au / \sqrt{\lambda}$ are mutually inverse bijections between the (one-dimensional) λ -eigenspaces, and they preserve the ordering of the shared nonzero eigenvalues. $\square$

Lemma 2 (Principal direction factor coordinates). *With the notation established in Section 3, write $K^{(p)} := \Sigma_f^{1/2} G_B^{(p)} \Sigma_f^{1/2}$ for $G_B^{(p)} := B^\top B / p$:*

1. *For large enough p , $K^{(p)}$ has distinct positive eigenvalues $\lambda_1^{(p)} > \dots > \lambda_k^{(p)} > 0$. Given the spectral decomposition $\Sigma_0 := B \Sigma_f B^\top = b \Delta_0 b^\top$ with $\Delta_0 := p \text{diag}(\lambda_1^{(p)}, \dots, \lambda_k^{(p)})$ in decreasing*

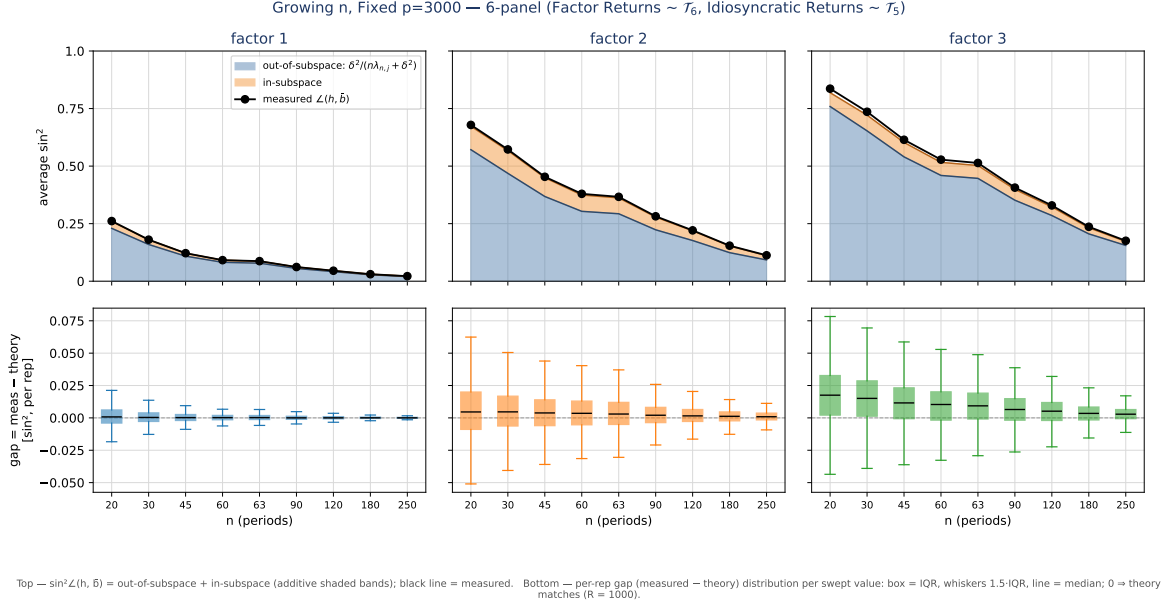


Figure 2: Errors between true and estimated factor exposures for $p = 3000$ variables based on n observations. Top panel: Black dots are finite p errors averaged over 1000 paths. Blue and orange bars are average out-of-subspace and in-subspace asymptotic limits. Bottom panel: Path-by-path differences between the $p = 3000$ angle between the true and estimated normalized factor exposures and the asymptotic limits.

order, there is a unique $k \times k$ orthogonal matrix $V^{(p)}$, with columns the unit eigenvectors of $K^{(p)}$, signed so that

$$\Sigma_f^{1/2} B^\top b = V^{(p)} \Delta_0^{1/2}. \quad (38)$$

(The exponent $1/2$ denotes the symmetric positive definite square root.)

Thus $K^{(p)} = V^{(p)} \text{diag}(\lambda_1^{(p)}, \dots, \lambda_k^{(p)}) V^{(p)\top}$ is its eigendecomposition.

2. Setting

$$C := \Sigma_f^{1/2} V^{(p)} \Delta_0^{-1/2}, \quad (39)$$

we have $BC = b$.

3. For $\phi := C^{-1}f$, we have

$$E[\phi \phi^\top] = C^{-1} \Sigma_f (C^{-1})^\top = \Delta_0. \quad (40)$$

Proof. Since $G_B^{(p)} \rightarrow G_B$ (Assumption 3), $K^{(p)} \rightarrow K := \Sigma_f^{1/2} G_B \Sigma_f^{1/2}$, whose eigenvalues are distinct by Assumption 4; hence, for large enough p , $K^{(p)}$ also has distinct eigenvalues $\lambda_1^{(p)} > \dots > \lambda_k^{(p)} > 0$, with orthonormal eigenvectors unique up to sign.

Set $A := B \Sigma_f^{1/2} \in \mathbb{R}^{p \times k}$, so $AA^\top = \Sigma_0$ and $A^\top A = \Sigma_f^{1/2} B^\top B \Sigma_f^{1/2} = pK^{(p)}$, which has simple nonzero eigenvalues $p\lambda_1^{(p)} > \dots > p\lambda_k^{(p)} > 0$. By Lemma 1, $\Sigma_0 = AA^\top$ has the same simple nonzero eigenvalues, giving the spectral decomposition $\Sigma_0 = b \Delta_0 b^\top$ with $\Delta_0 = p \text{diag}(\lambda_1^{(p)}, \dots, \lambda_k^{(p)})$, unique up to signs of b 's columns.

Still by Lemma 1, if b_j is the j th unit eigenvector of $\Sigma_0 = AA^\top$, then $v_j := A^\top b_j / \sqrt{p\lambda_j^{(p)}}$ is the j th unit eigenvector of $A^\top A = pK^{(p)}$, equivalently of $K^{(p)}$, since scaling by p does not change eigenvectors. Collecting $V^{(p)} := [v_1, \dots, v_k]$, orthonormality of the v_j (Lemma 1) makes $V^{(p)}$ orthogonal, and

$$\Sigma_f^{1/2} B^\top b = A^\top b = V^{(p)} \text{diag}(\sqrt{p\lambda_1^{(p)}}, \dots, \sqrt{p\lambda_k^{(p)}}) = V^{(p)} \Delta_0^{1/2}, \quad (41)$$

establishing the first part, with the signs of $V^{(p)}$'s columns tied to those of b .

For part 2, the display above gives $V^{(p)} \Delta_0^{-1/2} = A^\top b \Delta_0^{-1}$, so

$$BC = B \Sigma_f^{1/2} V^{(p)} \Delta_0^{-1/2} = A V^{(p)} \Delta_0^{-1/2} = A A^\top b \Delta_0^{-1} = \Sigma_0 b \Delta_0^{-1} = b \Delta_0 (b^\top b) \Delta_0^{-1} = b, \quad (42)$$

using $b^\top b = I_k$.

For part 3,

$$E[\phi \phi^\top] = C^{-1} E[ff^\top] (C^{-1})^\top = C^{-1} \Sigma_f (C^{-1})^\top = \Delta_0^{1/2} V^{(p)\top} \Sigma_f^{-1/2} \Sigma_f (\Sigma_f^{-1/2}) V^{(p)} \Delta_0^{1/2} = \Delta_0. \quad (43)$$

□

Lemma 3 (Noise concentration). *Write $Z_{\cdot l}$ for the l th column of Z , and let $a = a^{(p)} \in \mathbb{R}^p$ be deterministic unit vectors. Under Assumption 1, for each fixed l , $a^\top Z_{\cdot l} / \sqrt{p} \rightarrow 0$ almost surely under the conditional law given F , for almost every F .*

Proof. Work conditionally on F : by Assumption 1 the entries Z_{il} , $i \geq 1$, of the l th column are independent and mean zero, with variances $\delta_{i,l}^2$ and fourth moments $\mathbb{E}[Z_{il}^4 | F] \leq \kappa_4$; by the F -freeness of the variances and the uniform fourth-moment bound, all bounds below are deterministic. $A_p := a^\top Z_{\cdot l} = \sum_i a_i Z_{il}$ is a sum of (conditionally) independent mean-zero terms with, by $\|a\| = 1$, $\mathbb{E}[A_p^2 | F] = \sum_i a_i^2 \delta_{i,l}^2 \leq \bar{\delta}^2$ and $\mathbb{E}[A_p^4 | F] = \sum_i a_i^4 \mathbb{E}[Z_{il}^4 | F] + 3 \sum_{i \neq i'} a_i^2 a_{i'}^2 \delta_{i,l}^2 \delta_{i',l}^2 \leq \kappa_4 + 3\bar{\delta}^4 =: K$, using $\sum_i a_i^4 \leq (\sum_i a_i^2)^2 = 1$ (the cross terms with an odd power vanish by conditional independence and mean zero). By Markov, $P(|A_p| > \epsilon \sqrt{p} | F) = P(A_p^4 > \epsilon^4 p^2 | F) \leq K/(\epsilon^4 p^2)$, which is summable; Borel–Cantelli, applied under the conditional law, gives $\limsup_p |A_p|/\sqrt{p} \leq \epsilon$ a.s. for every $\epsilon > 0$. □

Lemma 4 (Eigenpair convergence). *Let symmetric $A^{(p)} \in \mathbb{R}^{m \times m}$ converge entrywise to A . If μ is a simple eigenvalue of A with unit eigenvector v , then for large p , $A^{(p)}$ has a simple eigenvalue $\mu^{(p)} \rightarrow \mu$ with unit eigenvector $v^{(p)} \rightarrow v$ (sign fixed by $\langle v^{(p)}, v \rangle \geq 0$).*

Proof. Weyl's inequality gives $|\lambda_i(A^{(p)}) - \lambda_i(A)| \leq \|A^{(p)} - A\| \rightarrow 0$, so the eigenvalue $\mu^{(p)}$ nearest μ is eventually the unique one within half the spectral gap of μ , is simple, and $\mu^{(p)} \rightarrow \mu$. The $v^{(p)}$ lie on the unit sphere; any subsequential limit $\bar{v}$ satisfies $A\bar{v} = \lim A^{(p)} v^{(p)} = \lim \mu^{(p)} v^{(p)} = \mu \bar{v}$ and $\langle \bar{v}, v \rangle \geq 0$, so by simplicity $\bar{v} = v$. Every subsequence thus has a further subsequence converging to v , whence $v^{(p)} \rightarrow v$. □

Proposition 1 (Dual limit). *Under Assumptions 1–3 and 5, conditional on F and a.s. as $p \rightarrow \infty$:*

- (a) $Z^\top Z / (np) \rightarrow (\delta^2/n) I_n$ in spectral norm;
- (b) $\|\Pi Z\|_F = \|b^\top Z\|_F = o(\sqrt{p})$;
- (c) $W^{(p,n)} \rightarrow W^{(n)}$ of (10) in spectral norm, with top k eigenpairs $(\lambda_j^{(n)} + \delta^2/n, w_j^{(n)})$; and
- (d) $\theta_j^{(p,n)} \rightarrow \lambda_j^{(n)} + \delta^2/n > 0$ and $w_j^{(p,n)} \rightarrow w_j^{(n)}$, $j = 1, \dots, k$.

Proof. (a) As n is fixed, entrywise a.s. convergence implies spectral-norm convergence. For each l , the diagonal entry,

$$\frac{1}{np} \sum_i Z_{il}^2 \rightarrow \delta^2/n$$

a.s. by Kolmogorov's strong law applied to the fixed sequence $\{Z_{il}^2\}_{i \geq 1}$. This is because, by our nesting convention, the law of each summand is unchanged as p grows, and the Z_{il}^2 are independent, conditionally on F , with $\text{Var} \leq \mathbb{E}[Z_{il}^4 | F] \leq \kappa_4$, so $\sum_i \text{Var}(Z_{il}^2 | F)/i^2 < \infty$, together with $\frac{1}{p} \sum_i \delta_{i,l}^2 \rightarrow \delta^2$.

For an off-diagonal entry ($l \neq m$), $\frac{1}{np} \sum_i Z_{il} Z_{im} \rightarrow 0$ a.s. likewise: conditionally on F the products $Z_{il} Z_{im}$, $i \geq 1$, are independent and mean zero (the entries of the array are mutually independent given F , by Assumption 1), with variance $\mathbb{E}[Z_{il}^2 | F] \mathbb{E}[Z_{im}^2 | F] = \delta_{i,l}^2 \delta_{i,m}^2 \leq \kappa_4$ by the Jensen bound of Assumption 1. Hence the matrix limit is $(\delta^2/n)I_n$.

(b) The (l, m) entry of $b^\top Z$ is $b_l^\top Z_{\cdot m}$ with b_l a deterministic unit vector, so $b_l^\top Z_{\cdot m} = o(\sqrt{p})$ a.s. by Lemma 3; summing the kn entries gives $\|b^\top Z\|_F = o(\sqrt{p})$. Since b has orthonormal columns, $\|\Pi Z\|_F = \|bb^\top Z\|_F = \|b^\top Z\|_F$.

(c) Substituting $Y = b\Phi + Z$ and using $b^\top b = I$,

$$W^{(p,n)} = \frac{\Phi^\top \Phi}{np} + \frac{\Phi^\top b^\top Z + Z^\top b\Phi}{np} + \frac{Z^\top Z}{np}. \quad (44)$$

By Lemma 5, the signal block $\Phi^\top \Phi/(np) = F^\top (B^\top B/p) F/n \rightarrow F^\top G_B F/n = W^{(n),0}$. The cross block is bounded in norm by $2\|\Phi\| \|b^\top Z\|_F/(np) \leq 2O(\sqrt{p}) o(\sqrt{p})/(np) = o(1)$, using $\|\Phi\| = \sqrt{p} \|\bar{\Phi}\| = O(\sqrt{p})$ and (b); and the noise block converges by (a). Hence $W^{(p,n)} \rightarrow W^{(n)}$. The eigenstructure of $W^{(n)}$ is read off in (10): $W^{(n),0}$ has rank k with eigenpairs $(\lambda_j^{(n)}, w_j^{(n)})$ (Assumption 5), and the scalar shift $(\delta^2/n)I_n$ moves eigenvalues by δ^2/n while fixing eigenvectors.

(d) The top k eigenvalues of $W^{(n)}$ are simple (Assumption 5); apply Lemma 4 on the a.s. event of (c). The limits are positive since $\lambda_j^{(n)} + \delta^2/n \geq \lambda_k^{(n)} + \delta^2/n > 0$. $\square$

Lemma 5 (Principal-coordinate convergence). *Write $G_B^{(p)} := B^\top B/p$. Fix an eigenbasis V of $K = \Sigma_f^{1/2} G_B \Sigma_f^{1/2}$, columns in decreasing eigenvalue order (unique up to column signs, by Assumption 4). Choose the column signs of b (free in Lemma 2) so that $\langle V_j^{(p)}, V_j \rangle \geq 0$ for each j and all large p . All limits below, and their uses in the proof of Theorem 1, are stated under this single convention; no conclusion of Theorem 1 depends on it. Then, under our standing assumptions,*

(i) $\Phi^\top \Phi/(np) = F^\top G_B^{(p)} F/n \rightarrow F^\top G_B F/n = W^{(n),0}$;

(ii) $\bar{\Phi} := \Phi/\sqrt{p} \rightarrow \bar{\Phi}^\infty$, where

$$\bar{\Phi}^\infty := \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_k}) V^\top \Sigma_f^{-1/2} F \quad (45)$$

and V is the eigenbasis of $\Sigma_f^{1/2} G_B \Sigma_f^{1/2}$; hence $N^{(p,n)} = \bar{\Phi} \bar{\Phi}^\top /n \rightarrow N^{(n)} := \bar{\Phi}^\infty (\bar{\Phi}^\infty)^\top /n$, with $\lambda_j^{(p)} \rightarrow \lambda_j$; explicitly, for each fixed n ,

$$N^{(n)} = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_k}) V^\top \Sigma_f^{-1/2} \frac{F F^\top}{n} \Sigma_f^{-1/2} V \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_k}); \quad (46)$$

and

(iii) $N^{(n)}$ has the same nonzero spectrum $\{\lambda_j^{(n)}\}$ as $W^{(n),0}$, with j th unit eigenvector $\nu_j^{(n)}$ satisfying

$$\bar{\Phi}^\infty w_j^{(n)} = \sqrt{n\lambda_j^{(n)}} \nu_j^{(n)}.$$

(iv) In addition, dropping the fixed- F conditioning of Section 4, but assuming the factor sample $f^{(1)}, f^{(2)}, \dots$ satisfies $FF^\top/n \rightarrow \Sigma_f$ a.s. as $n \rightarrow \infty$ (see the proof for two standard sufficient conditions for this hypothesis),

$$N^{(n)} \rightarrow \text{diag}(\lambda_1, \dots, \lambda_k)$$

almost surely as $n \rightarrow \infty$, so $\nu_j^{(n)} \rightarrow e_j$.

Part (iv) is used only to interpret the limits of the fixed- n theory.

Proof. (i) By Lemma 2, $\Phi = (C^{(p)})^{-1}F = \Delta_0^{1/2}V^{(p)\top}\Sigma_f^{-1/2}F$, so

$$\Phi^\top\Phi = F^\top\Sigma_f^{-1/2}V^{(p)}\Delta_0V^{(p)\top}\Sigma_f^{-1/2}F.$$

Since $K^{(p)} = V^{(p)}\text{diag}(\lambda_1^{(p)}, \dots, \lambda_k^{(p)})V^{(p)\top}$ (Lemma 2) and $\Delta_0 = p\text{diag}(\lambda_1^{(p)}, \dots, \lambda_k^{(p)})$,

$$V^{(p)}\Delta_0V^{(p)\top} = pK^{(p)} = \Sigma_f^{1/2}B^\top B\Sigma_f^{1/2},$$

whence $\Phi^\top\Phi = F^\top B^\top BF$ and $\Phi^\top\Phi/(np) = F^\top G_B^{(p)}F/n$. Assumption 3 gives the limit.

(ii) By Lemma 2, $K^{(p)} = V^{(p)}\text{diag}(\lambda_1^{(p)}, \dots, \lambda_k^{(p)})V^{(p)\top}$ is its eigendecomposition, and (as shown in the proof of Lemma 2) $K^{(p)} \rightarrow K := \Sigma_f^{1/2}G_B\Sigma_f^{1/2}$, whose eigenvalues $\lambda_1 > \dots > \lambda_k > 0$ are simple (Assumption 4). Lemma 4 gives $V^{(p)} \rightarrow V$ under the sign convention adopted in the statement, and $\lambda_j^{(p)} \rightarrow \lambda_j$; with $\bar{\Phi} = \text{diag}(\sqrt{\lambda_j^{(p)}})V^{(p)\top}\Sigma_f^{-1/2}F$ (from $\Phi = \Delta_0^{1/2}V^{(p)\top}\Sigma_f^{-1/2}F$ and $\Delta_0^{1/2}/\sqrt{p} = \text{diag}(\sqrt{\lambda_j^{(p)}})$) this yields $\bar{\Phi} \rightarrow \bar{\Phi}^\infty$ and $N^{(p,n)} \rightarrow N^{(n)}$. Substituting the expression for $\bar{\Phi}^\infty$ into $N^{(n)} = \bar{\Phi}^\infty(\bar{\Phi}^\infty)^\top/n$ gives (46).

(iii) Since $V\text{diag}(\lambda_j)V^\top = K = \Sigma_f^{1/2}G_B\Sigma_f^{1/2}$,

$$\frac{(\bar{\Phi}^\infty)^\top\bar{\Phi}^\infty}{n} = \frac{F^\top\Sigma_f^{-1/2}V\text{diag}(\lambda_j)V^\top\Sigma_f^{-1/2}F}{n} = \frac{F^\top\Sigma_f^{-1/2}\Sigma_f^{1/2}G_B\Sigma_f^{1/2}\Sigma_f^{-1/2}F}{n} = \frac{F^\top G_B F}{n} = W^{(n),0}.$$

Thus $N^{(n)} = \bar{\Phi}^\infty(\bar{\Phi}^\infty)^\top/n$ and $W^{(n),0}$ are Gram duals, sharing the spectrum $\{\lambda_j^{(n)}\}$; Lemma 1 applied to $\bar{\Phi}^\infty/\sqrt{n}$ gives the j th eigenvector of $N^{(n)}$ as $\nu_j^{(n)} = \bar{\Phi}^\infty w_j^{(n)}/\sqrt{n\lambda_j^{(n)}}$.

(iv) Under the hypothesis $FF^\top/n \rightarrow \Sigma_f$ a.s., the closed form (46) gives

$$N^{(n)} \rightarrow \text{diag}(\sqrt{\lambda_j})V^\top V\text{diag}(\sqrt{\lambda_j}) = \text{diag}(\lambda_j).$$

The limit's eigenvalues λ_j are simple (Assumption 4) with eigenvectors the axes e_j , so Lemma 4 gives $\nu_j^{(n)} \rightarrow e_j$.

We give two sufficient conditions for the hypothesis. Since $FF^\top/n = \frac{1}{n}\sum_{l=1}^n f^{(l)}f^{(l)\top}$ and k is fixed, entrywise a.s. convergence suffices; fix an entry (a, b) and set $X_l := f_a^{(l)}f_b^{(l)}$.

Independent factors. Suppose the $f^{(l)}$ are independent with common second-moment matrix $\mathbb{E}[f^{(l)}f^{(l)\top}] = \Sigma_f$ and uniformly bounded fourth moments $\kappa_f := \sup_l \mathbb{E}\|f^{(l)}\|^4 < \infty$. Then the X_l are independent with common mean $\mathbb{E}X_l = (\Sigma_f)_{ab}$ and, by Cauchy–Schwarz,

$$\text{Var}(X_l) \leq \mathbb{E}[(f_a^{(l)})^2(f_b^{(l)})^2] \leq (\mathbb{E}[(f_a^{(l)})^4]\mathbb{E}[(f_b^{(l)})^4])^{1/2} \leq \kappa_f < \infty,$$

so $\sum_l \text{Var}(X_l)/l^2 < \infty$ and Kolmogorov's strong law for independent (not necessarily identically distributed) summands gives $\frac{1}{n} \sum_{l \leq n} X_l \rightarrow (\Sigma_f)_{ab}$ a.s.

Stationary ergodic factors. Suppose $f^{(1)}, f^{(2)}, \dots$ is stationary and ergodic with $\mathbb{E}\|f^{(1)}\|^2 < \infty$ and $\mathbb{E}[f^{(1)} f^{(1)\top}] = \Sigma_f$. Then $(X_l)_{l \geq 1}$, a fixed measurable function of the shifted path, is stationary and ergodic with $\mathbb{E}|X_l| \leq \mathbb{E}\|f^{(1)}\|^2 < \infty$, and Birkhoff's ergodic theorem gives $\frac{1}{n} \sum_{l \leq n} X_l \rightarrow \mathbb{E}X_1 = (\Sigma_f)_{ab}$ a.s. This covers serially dependent factor paths, and requires only second moments. $\square$

B Symbol Tables

Object	Truth	Rotated truth (principal coordinates)	Estimated (sample)
Loadings / directions	$B \in \mathbb{R}^{p \times k}$, rank k , not separately identifiable	$b = BC$, orthonormal columns b_j : the principal directions of Σ_0 (estimation targets)	$h_j = h_j^{(p,n)}$, top- k unit eigenvectors of $S^{(p,n)}$; collected as $H \in \mathbb{R}^{p \times k}$
Signal (1 obs.)	$f \in \mathbb{R}^k$, $\mathbb{E}[ff^\top] = \Sigma_f$	$\phi = C^{-1}f$, $\mathbb{E}[\phi\phi^\top] = \Delta_0$ (diagonal)	—
Signal (n obs.)	$F = [f^{(1)} \dots f^{(n)}] \in \mathbb{R}^{k \times n}$ (p -free)	$\Phi = C^{-1}F$ (p -dependent via C); scaled $\bar{\Phi} = \Phi/\sqrt{p} \rightarrow \bar{\Phi}_\infty$	—
Noise	$z \in \mathbb{R}^p$; $Z \in \mathbb{R}^{p \times n}$	same Z (the rotation acts only on the signal term)	enters limits via $Z^\top Z/p \rightarrow \delta^2 I_n$
Data	$Y = BF + Z$	$Y = b\Phi + Z$ (same matrix, refactored)	Y is observed
Covariance ($p \times p$)	$\Sigma_y = \Sigma_0 + \Delta_z$, $\Sigma_0 = B\Sigma_f B^\top$	$\Sigma_0 = b\Delta_0 b^\top$; scaled eigenvalues $\lambda_j^{(p)} \rightarrow \lambda_j$	$S^{(p,n)} = YY^\top/(np)$, eigenvalues $\theta_j^{(p,n)}$
$n \times n$ dual	—	$W^{(n),0} = F^\top G_B F/n$, eigenpairs $(\lambda_j^{(n)}, w_j^{(n)})$	$W^{(p,n)} = Y^\top Y/(np)$, eigenpairs $(\theta_j^{(p,n)}, w_j^{(p,n)})$; limit $W^{(n)} = W^{(n),0} + (\delta^2/n)I_n$
$k \times k$ signal Gram	$K = \Sigma_f^{1/2} G_B \Sigma_f^{1/2}$, eigenvalues λ_j	$N^{(p,n)} = \Phi\Phi^\top/(np) \rightarrow N^{(n)}$, eigenpairs $(\lambda_j^{(n)}, \nu_j^{(n)})$	— (realized, not estimated; $\nu_j^{(n)}$ rotates off the axis e_j)
Target in coordinates	—	$b_j \leftrightarrow e_j$ via the isometry $\Gamma(\xi) = b\xi$	coordinate of Πh_j : $\xi_j^{(p,n)} \rightarrow \kappa_j^{(n)} \nu_j^{(n)}$

Table 3: Truth, rotated-truth, and sample objects. The rotation C of Lemma 2 connects (B, F) to (b, Φ) ; the noise Z and data Y are common to both factorizations. Three layers of eigenstructure appear: population (λ_j, b_j) , realized at finite n $(\lambda_j^{(n)}, \nu_j^{(n)})$, and sample $(\theta_j^{(p,n)}, h_j)$.

Matrix	Definition	Size	Eigenvalues	Eigenvectors
<i>Ambient p-dimensional covariances</i>				
Σ_0/p	$B\Sigma_f B^\top/p$ (rank k)	$p \times p$	$\lambda_j^{(p)}$	b_j
$S^{(p,n)}$	$YY^\top/(np)$	$p \times p$	$\theta_j^{(p,n)}$	$h_j = h_j^{(p,n)}$
$S^{(p,n),0}$	$b\Phi\Phi^\top b^\top/(np)$ (rank k , noiseless)	$p \times p$	$\lambda_j^{(p,n)}$	$h_j^{(p,n),0}$
<i>$n \times n$ duals (same nonzero spectrum as the block above)</i>				
$W^{(p,n)}$	$Y^\top Y/(np)$	$n \times n$	$\theta_j^{(p,n)}$	$w_j^{(p,n)}$
$W^{(n)}$	$F^\top G_B F/n + (\delta^2/n)I_n$ ($= \lim_p W^{(p,n)}$)	$n \times n$	$\lambda_j^{(n)} + \delta^2/n$	$w_j^{(n)}$
$W^{(p,n),0}$	$\Phi^\top \Phi/(np) = F^\top G_B^{(p)} F/n$ (noiseless dual)	$n \times n$	$\lambda_j^{(p,n)}$	$w_j^{(p,n),0}$
$W^{(n),0}$	$F^\top G_B F/n$ (noiseless limit)	$n \times n$	$\lambda_j^{(n)}$	$w_j^{(n)}$
<i>$k \times k$ signal duals</i>				
$N^{(p,n)}$	$\Phi\Phi^\top/(np)$	$k \times k$	$\lambda_j^{(p,n)}$	$\nu_j^{(p,n)}$
$N^{(n)}$	$\bar{\Phi}^\infty(\bar{\Phi}^\infty)^\top/n$ ($= \lim_p N^{(p,n)}$)	$k \times k$	$\lambda_j^{(n)}$	$\nu_j^{(n)}$
$K^{(p)}$	$\Sigma_f^{1/2} G_B^{(p)} \Sigma_f^{1/2}$ (dual of Σ_0/p ; eigenbasis $V^{(p)}$, Lemma 2)	$k \times k$	$\lambda_j^{(p)}$	—
K	$\Sigma_f^{1/2} G_B \Sigma_f^{1/2}$ ($= \lim_p K^{(p)}$)	$k \times k$	λ_j	—
<i>$k \times k$ loading Gram</i>				
$G_B^{(p)}$	$B^\top B/p$	$k \times k$	> 0	—
G_B	$\lim_p B^\top B/p$	$k \times k$	> 0	—

Table 4: Matrices appearing in the proofs of Theorem 1 and Corollary 3, with their eigenvalue and eigenvector labels. Throughout $j = 1, \dots, k$ indexes the top eigenpairs in decreasing order. Within each grouped block the matrices share the nonzero spectrum in the corresponding column (Gram duality, Lemma 1): the noisy sample pair $S^{(p,n)}, W^{(p,n)}$ share $\theta_j^{(p,n)}$; the noiseless sample triad $S^{(p,n),0}, W^{(p,n),0}, N^{(p,n)}$ share $\lambda_j^{(p,n)}$; and the noiseless limits $W^{(n),0}, N^{(n)}$ share $\lambda_j^{(n)}$, while Σ_0/p and $K^{(p)}$ share the population spectrum $\lambda_j^{(p)} \rightarrow \lambda_j$ (Lemma 2).

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